2.4.5 Class: Chondrichthyes

(*G. chondros* –cartilage; *ichthys* -fish)

They are marine fishes with cartilaginous endoskeleton. Notochord is persistent throughout life. Skin is tough covered by dermal **placoid scales** and the caudal fin is **heterocercal** (asymmetrical both externally and internally). Mouth is located ventrally and teeth are modified placoid scales which are backwardly directed. Their jaws are very powerful and are predaceous animals. Respiration by **lamelliform gills** without operculum (gill cover). Excretory organs are **mesonephric** kidneys. Two chambered heart is present. Cartilaginous fishes are **ureotelic** and store urea in their blood to maintain osmotic concentration of body fluids. They are **poikilothermic** and **viviparous**. Sexes are separate. In males pelvic fins bear claspers to aid in internal fertilisation.

Examples: *Scoliodon* (Shark), *Trygon* (Sting ray), *Pristis* (Saw fish) (Figure 2.23).

2.4.6 Class: Osteichthyes

(G. *osteon* –bone; *ichthys* -fish)

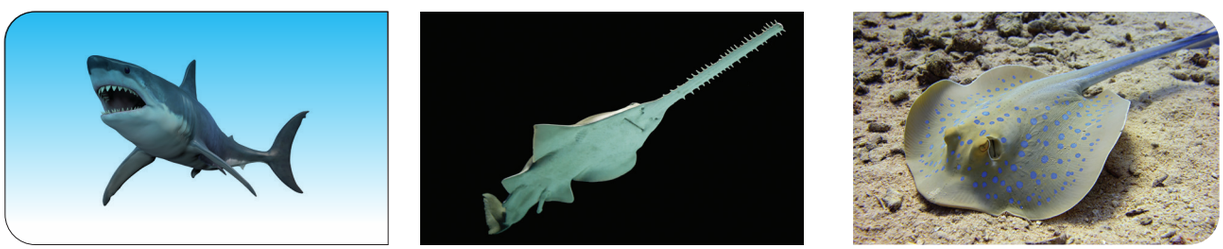
It includes both marine and freshwater fishes with bony endoskeleton and spindle shaped body. Skin is covered by **ganoid**, **cycloid** or **ctenoid scales**. Respiration is by four pairs of filamentous gills and is covered by an operculum on either side. Air bladder is present with or without a connection to the gut. It helps in gaseous exchange (lung fishes) and for maintaining buoyancy in most of the ray finned fishes. They have a ventrally placed two chambered heart. Excretory organs are **mesonephric kidneys** and are **ammonotelic**. Presence of well developed **lateral line sense organ**. Sexes are separate, external fertilization is seen and most forms are oviparous (Figure 2.24).

Examples: *Exocoetus* (Flying fish), *Hippocampus* (Sea horse), *Labeo* (Rohu), *Catla* (Catla), *Echeneis* (Sucker fish), *Pterophyllum* (Angel fish)

2.4.7 Class: Amphibia

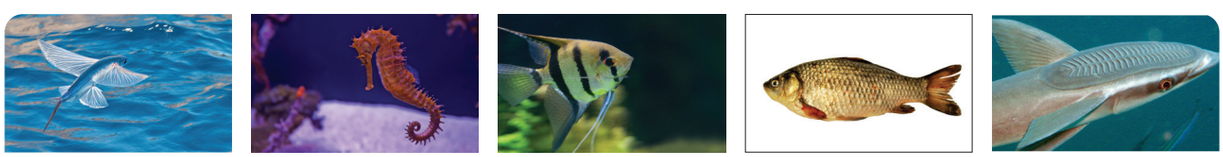
(G. *amphi*-both; *bios* -life)

Amphibians are the first vertebrates and tetrapods to live both in aquatic as well as terrestrial habitats.They are **poikilothermic**. Their body is divisible into the head and trunk and most of them have two pairs of limbs; tail may or may not be present. Their



Shark *Pristis* *Sting ray*

Figure 2.23 Examples of Chondrichthyes



Flying fish *Sea horse* *Angel fish* *Carp* *Sucker fish*

Figure 2.24 Examples of Osteichthyes

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